

MountMonitor

v1.20.0

User Manual

Real-time telescope mount monitoring
for astrophotography

Theory · Installation · Use · Reference

After MountMonitor v3.37 by Nicolàs de Hilster
Python/PyQt6 rewrite

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Part I

Understanding

Chapter 1

Celestial coordinates

1.1 The celestial sphere

Seen from the ground, the night sky behaves like a sphere turning from east to west about an axis through the celestial poles. Stars lie at wildly different distances, but their position on that fictitious sphere is all one needs to find and follow them.

The Earth completes one turn in **23 h 56 min 4 s** — the *sidereal day*, shorter than the 24 hours of the solar day. That, and not the clock, is the rate a mount must turn at.

1.2 Right ascension

Right ascension (RA) is the celestial equivalent of longitude. It is counted in **hours** rather than degrees, from 0 h to 24 h, starting at the vernal point.

Formula

$$1 \text{ h} = 15^\circ \quad 1 \text{ min} = 15' \quad 1 \text{ s} = 15''$$

$$\text{So RA} = 02 \text{ h } 44 \text{ min } 08 \text{ s is } 2 \times 15^\circ + 44 \times 0.25^\circ + 8 \times \frac{15}{3600}^\circ = 41.03^\circ.$$

i Information

RA is the critical coordinate for tracking: a one-second timing error is worth 15 arcseconds on the sky at the celestial equator. At $1.6''/\text{pixel}$ that is about ten pixels of trailing.

1.3 Declination

Declination (DEC) is the equivalent of latitude: the angular distance north or south of the celestial equator, from -90° to $+90^\circ$. On a correctly aligned mount the declination axis does not move while tracking.

1.4 Hour angle

The **hour angle** measures how far an object stands from the local meridian. It grows continuously as the Earth turns.

Formula

$$\text{HA} = \text{local sidereal time} - \text{RA}$$

An RA tolerance can be expressed in arcseconds or in seconds of time. The conversion depends on declination:

$$\text{seconds of time} = \frac{\text{arcseconds}}{15 \cos(\text{DEC})}$$

✓ Tip

Near the celestial pole an RA error barely moves the star on the sensor: the cosine of the declination damps it. Circumpolar objects are therefore more forgiving of RA tracking errors — but not of declination ones.

Chapter 2

Equatorial mounts

2.1 Principle

An **equatorial mount** cancels the Earth's rotation by turning the tube about an axis parallel to the Earth's own: the polar axis.

- **Polar axis (RA)**: turns at the sidereal rate.
- **Declination axis**: still while tracking, in theory.

Formula

$$\omega = \frac{1\,296\,000''}{86\,164.09\,\text{s}} \approx 15.041 \text{ arcseconds per second}$$

Any departure from that rate, and any misalignment of the polar axis, becomes a drift the image records.

2.2 10Micron mounts

10Micron mounts (GM1000 HPS and above) carry **absolute encoders**: they know where they point even after a power cut, where a relative encoder loses it. To that they add a worm drive, periodic-error correction and, above all, an internal **pointing model** built from several dozen stars.

That model is what makes **unguided** imaging possible: the mount does not merely turn at the sidereal rate, it applies corrected rates that absorb flexure, refraction and the residual polar misalignment.

Warning

A direct consequence, often misunderstood: on a modelled mount the **polar alignment error lives inside the model** and is already compensated by the tracking rates. A residual declination drift therefore does not call for the azimuth screw, but for **more model points near the target**.

These mounts speak an **extended LX200**: beyond the standard commands they expose raw axis positions, applied rates, model state and environmental data. MountMonitor uses those extensions where available.

Chapter 3

What degrades tracking

No mount tracks perfectly. The causes overlap, and each leaves a different signature in the record.

3.1 Periodic error

A worm is never perfectly round nor perfectly centred. Its defect repeats with every turn and produces an oscillation of fixed period — a few minutes, most often. That is exactly what a Fourier analysis can pull out (chapter ??).

3.2 Polar alignment

If the polar axis does not point exactly at the celestial pole, the star drifts slowly in declination, the faster the further one is from the meridian. On a mount without a model the answer is mechanical. On a modelled mount, see the warning in the previous chapter.

3.3 Atmospheric refraction

The atmosphere lifts objects the more they are low: some 34' at the horizon, 1' at 45° of altitude. Refraction therefore changes through the night, and its drift is easily mistaken for a misalignment.

3.4 Mechanical disturbance and vibration

Wind, a cable pulling, ground movement, a meridian flip: so many one-off jolts, none of them periodic. They show as isolated spikes, and that is what tells them apart.

3.5 Commanded moves

These are not errors at all. Between exposures the sequencer deliberately moves the mount — **dither** to decorrelate noise, **re-centering** when the target has slipped — and the mount *stays* at its new position. Confusing them with tracking error is the costliest mistake one can make reading a record, and chapter ?? is devoted to it.

Chapter 4

Reading the numbers

Warning

This is the most important chapter in the manual. A tracking record is read wrongly with disconcerting ease, and the figure that comes out is then wrong by a factor of several hundred.

4.1 The signal is a staircase

Here is what a night of unguided imaging actually records: the mount holds its position to a fraction of an arcsecond, then the sequencer dithers it, and it holds its new position just as well. The trace is not a noisy line about a mean: it is a **staircase**.

A standard deviation taken over the whole night then measures the height of the steps, not the quality of the tracking. And removing a straight line — what a classic detrending does — does not remove a staircase.

Information

On a real eight-and-a-half hour session the raw combined RMS came out at **86.3''**, a damning verdict. The mount was in fact holding each position to **0.18''**. The two figures differ by a factor of 480.

4.2 Jitter

Jitter is the spread the mount shows *while* it holds a position. It is the only figure that smears a star on the sensor, and the one MountMonitor bases its rating on.

It is measured between repositionings, never about a straight line. The program cuts the record at the steps, drops two samples after each — the move itself is not tracking — and takes the median of the per-step spreads.

Tip

This approach has a property worth noting: a sequence that dithers on every frame does not come out worse than one that never does. The figure describes the mount, not the sequencer.

4.3 Commanded moves

They are counted and classified separately:

Class	What it is
Dither	A small move between exposures, to decorrelate noise.
Re-centering	A larger move, bringing the target back where it belongs.
Unsettled	The spread that follows stays several times the usual value. That is the one worth looking at.

Warning

A threshold crossing is not a move. A 70" re-centering takes several samples to complete and trips the threshold at each one: one night read as 98 crossings for 53 actual moves, with a median amplitude ten times too small. MountMonitor groups crossings less than ten seconds apart.

4.4 Slow drift

It is quoted per hour and per exposure. On an unguided mount every dither cancels it: what matters is not the drift over the night, but the displacement it produces *during one exposure*.

Formula

$$\text{displacement over one exposure} = \text{drift (\"/h)} \times \frac{\text{exposure (s)}}{3600}$$

A drift of $-16.8''/\text{h}$ therefore moves the star by $0.84''$ over a 180 s exposure.

4.5 Running standard deviation

Computed live over a 60 to 900 second window, it serves to see *when* something happened, not to grade the night. After a target change, an acquisition gap or a repositioning, its window straddles the move and describes that rather than the tracking: those values are excluded from the report.

4.6 The rating

It rests on the combined jitter, with thresholds suited to the type of mount. An unguided 10Micron is judged on wider thresholds than a guided mount, because the orders of magnitude are not the same.

Rating	Guided mount	Unguided precision
Excellent	$< 0.5''$	$< 2.0''$
Good	$< 1.5''$	$< 4.0''$
Fair	$< 3.0''$	$< 8.0''$

Tip

Always relate the rating to your sampling. At $1.6''/\text{pixel}$, a $0.5''$ jitter shakes the star by a third of a pixel: invisible on the final image.

Chapter 5

Fourier analysis

5.1 What it is for

A Fourier transform breaks a signal into a sum of sinusoids. On a tracking record it brings out what *repeats* — and a worm periodic error repeats by construction.

MountMonitor looks in the **2 to 15 minute** band, where the worm periods of common mounts fall.

5.2 Two precautions

- The transform runs on **detrended** deviations. A linear drift is not periodic, but over a finite window it produces a large low-frequency component that would hide what is being looked for.
- The **Nyquist limit** requires sampling at least twice per period. At 2 Hz, periods below one second are out of reach — no real inconvenience, no mechanical error lives there.

5.3 Reading a spectrum

A sharp peak at a stable period marks a periodic error. A flat spectrum means there is no detectable one — the case of 10Micron mounts after correction. Broad, wandering peaks are noise, usually wind.